

What's available?

The burden of lifestyle and non-communicable diseases has raised the demand of medical electronics solutions which can be used at home for monitoring and detection purposes. This leads to a sharing of responsibility of your own health and shifting the gears from remedial action to preventive diagnostics. Innovative R&D in diagnostics has changed the way sensors were realised earlier. Presently, sensors have become smaller in size and work on smaller batteries for longer durations.

The next-level devices include monitoring and control. For example, insulin injectors monitor the blood sugar level and then inject appropriate medication volume in the body, forming a closed-loop system with monitor and control functionalities.

Portability is being introduced now in the surgical procedures too. More and more electronic devices (like cutters, reamers and drills, etc) used in surgery are being re-designed by many companies with portability in mind. Surgical procedures will then cross the barriers of access if portability can be introduced to a large extent in the operation theatres.

Teleradiology enables the transfer of acquired images from one diagnostic centre or hospital to anywhere across the world through the Internet where the caregiver can refer to it and suggest further course of treatment. Portable electrocardiogram (ECG), electromyography (EMG) and ultrasound machines offer a quick diagnostic solution in situations where it may not be possible to move the patient and to provide healthcare in villages and remote areas. With digital diagnostics, nowadays, patients do not need to carry X-ray or ultrasound reports to the doctor for diagnosis. All reports and test data can be encapsulated into digital media and can be accessed any-

where on a computer.

Battery life plays an integral role in the functionality and success of portable medical electronic devices. A two-way approach of developing batteries with longer lives as well as developing devices that consume less power will form a solution. Research is underway on these aspects and new products are being introduced in the market.

With the spread of mobile phones and their usage, these have become a target device for medical applications. In recent years, there has been a lot of focus on development of mobile applications for Android, Black-

Amalgamation of technology innovation and cost-effectiveness will lead to the desired results.

More functionality can be built in mobile phone medical applications with addition of sensors and embedded software for monitoring of vital parameters in cardiac patients. Any irregularity in the monitoring results can be raised as an alarm and then transmitted to the caregiver, who can suggest immediate measures for medication.

Medical healthcare solution providers are looking forward to health information technology (IT) and point-of-care therapy. Various applications will come up in the domains of electronic health record, electronic medical record, telemedicine and beyond, where medical electronics will serve as a base architecture for high-end network applications. The introduction of e-ICU has enabled the doctors to have a real-time monitoring of the patients in intensive

care unit while they are on the move or elsewhere.

Medical electronic devices in present and future will adhere to communication protocols, which can securely transfer data over wireless media/Internet and bridge the gap between caregiver and patient, also reducing the reaction time and hence increasing the opportunity. Data security and authentication will play a key role and extend the participation of IT and other allied software encryption services.

The future of medical healthcare's access and affordability will depend on the technological advancements in the domain of medical electronics. A collaborative development effort amongst various disciplines, centred on the medical electronic devices will expand the reach and benefits to a larger section of the Indian population. ●



Philips Lifeline with AutoAlert for seniors can detect falls and automatically call for help

berry, iPhone, etc.

Microelectromechanical systems (MEMS) and accelerometer-based solutions are embedded in a mobile phone and used for applications like pulse-rate and heartbeat monitoring of the user. The same data can also be transmitted to the caregiver.

However, a challenge lies in the regulation of these devices and software applications so that the data confirmed by these portable devices is validated and correct. Therefore a strong regulatory framework needs to be devised for promoting the right products.

Way forward

Medical electronics is a domain of very high rate of technology innovation and obsolescence at the same time. Product development and research has to be promoted with a framework for protection of intellectual property rights. Product development in this sunrise sector must take a collaborative path, with joint development across the nations and simultaneous global launch to offset the high R&D cost.

The FICCI MEF (FICCI Medical Electronics Forum) launched by Federation of Indian Chambers of Commerce & Industry (FICCI) responds to the requirements of the medical electronics industry and creates awareness about the opportunities as the next growth area after IT sector